

Faculty of Information Technology
Course: *Data Visualization using Tableau*
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Netflix Movies and TV Shows

The catalog as designed supply — a visualization study through Stanford Design Thinking and Munzner’s nested model

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Corpus	Flixable / Kaggle Netflix titles extract · $N = 8,807$ titles
Window	date_added 2008-01-01 → 2021-09-25 (2021 right-censored)
Prototype	Tableau workbook Tableau/Netflix & TV Show.twbx
Publication	Python pipeline <code>python -m netflix_catalog → figures/viz_*.png</code>
Document	Course report (academic rewrite). Dual evidence: left Tableau, right Python.



Working paper for the studio course. Unit of analysis is a title, not a viewer. This PDF supersedes the September 2025 Word draft (heart-disease copy-paste, rating-as-reviews, country-as-filming-location, and ranking “Không xác định” as a country are corrected).

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Abstract

A public catalog of titles is not a panel of viewers. This study treats the Flixable/Kaggle Netflix titles extract ($N = 8,807$; last `date_added` = 25 September 2021) as a **designed supply object**: the set of works a platform chose to license, produce, and surface. We use the Stanford d.school's five modes (Empathize, Define, Ideate, Prototype, Test) as the *research protocol* and Munzner's nested model (domain → data/task abstraction → idiom → algorithm) as the *visualization protocol*.

The interactive prototype is a Tableau workbook. The publication layer is a Python pipeline that keeps `NULL` in analytic tables, recodes three duration-in-`rating` rows, explodes multi-label country and genre credits at a documented grain, and redraws each course chart with sentence titles, source lines, and a palette sampled from the workbook (cream `#EFEFE8`, Movie `#E15658`, TV Show `#F28B25`).

Primary result. By 2021 the catalog is **mature-skewed and geographically concentrated**. Movies are 69.6% of titles (6,131 / 8,807). TV-MA + TV-14 account for 60.9% of titles. Family-adjacent ratings are 23.4%. Exploded production credits are headed by the United States (3,690), India (1,046), and the United Kingdom (806). If missing country is promoted to a nation label, **Unknown ranks third (831 credits, 9.4% of titles)** — which is why the Python ranking excludes it. Additions (`date_added`) peak in 2019 (2,016 titles); 2021 is truncated.

Keywords: catalog analytics, construct validity, missingness, unit of analysis, design thinking, nested model, Tableau, reproducibility.

How to read the dual evidence

Every domain task below is shown as **two panels**:

Column	Role
Left — Tableau	Course prototype: interactive, filterable, the object a meeting can point at. Some sheets still reflect the studio extract (Vietnamese <code>Không xác định</code> , duration strings in <code>rating</code>).
Right — Python	Publication figure: same <i>question</i> , corrected grain, sentence title, missingness handled, source line, Tableau-harmonious color.

A few views exist on only one side, and that is stated: missingness is a Python diagnostic the workbook never isolated; the choropleth is kept as Tableau because a map is the right idiom for geography.

Grammar on every result card:

1. **Observation** — what the chart shows at the stated grain
2. **Inference** — what that implies *about the catalog*
3. **Non-claim** — what a reader is not allowed to conclude

1. Introduction

1.1 Design challenge

Netflix's public identity is personalization. The file in this repository is the opposite of a recommendation log: it is a **title-level catalog snapshot**. Using it as an audience study produces the most common error in student (and industry) streaming analytics: **inferring preference from inventory**.

*How might a content, product, or regional team **see the catalog as a designed object** — its growth, geography, maturity mix, and genre architecture — without pretending the file contains viewers?*

That is an ill-structured problem. Stanford Design Thinking exists for those. Munzner's nested model exists so that a wrong *domain* question is not laundered by a pretty *idiom*.



Figure 1. Object of study: a global catalog. Subscriber counts, ARPU, and watch-time are out of sample.

1.2 Contributions

1. A **construct-correct dictionary**: `rating` is a parental-guideline / MPAA-style maturity label, not a crowd score; `country` is a production-credit field, not filming location; `date_added` is catalog-entry time, not cinematic release.
2. A **cleaning protocol that refuses cosmetic completeness**: analytic tables keep `NULL`; English `Unknown` exists only in the Tableau extract; `Unknown` is never ranked as a producing country.
3. A **dual prototype**: Tableau for interaction, Python for publication claims that can survive Test.
4. Explicit **non-claims** and threats to validity, which we treat as part of the result, not an appendix apology.

1.3 What this paper is not

It is not a viewer study, a cost study, or a 2022–2026 originals tracker. Claims are bound to this extract.

2. Related work and theoretical frame

Catalog versus demand. Media-industry research (e.g. Lotz on internet-distributed television) distinguishes *what a service carries* from *what a household watches*. This extract can speak to the former. Treating `listed_in` tag volume as “what teens prefer” is a construct error, not a small caveat.

Design thinking as protocol, not poster. The Hasso Plattner Institute of Design at Stanford (d.school, *Design Thinking Bootleg*, 2018) specifies five *modes* with return paths. Empathy findings reopen when a test fails. We use that loop as the spine of the analysis, not as a cover illustration.

Nested model for visualization. Munzner (2014) separates domain situation, data/task abstraction, visual encoding, and algorithm. A failure at a higher level cannot be rescued by a lower one: a heatmap named “correlations” that computes no coefficient is an abstraction failure, regardless of color scale. Mackinlay’s (1986) expressiveness/effectiveness criteria and Cleveland & McGill’s ranking of perceptual tasks (position $\gg$ length $\gg$ area $\gg$ color) govern idiom choice in Section 6.

Missingness as signal. Rubin’s taxonomy (MCAR / MAR / MNAR) is the reason we do not fill TV-show `direct - or` with a display string and then “profile completeness = 100%.” Director missingness is 91.4% for TV Shows versus 3.1% for Movies: a schema/practice difference, not a hole.

3. Research protocol (Stanford d.school)

The five modes are not a waterfall.

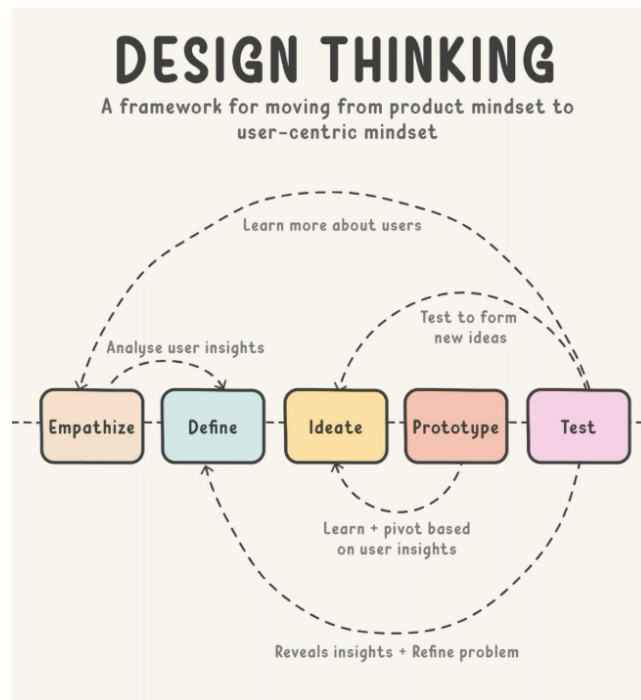


Figure 2. Iterative protocol used as the research spine.

3.1 Empathize

This file contains no user IDs, sessions, or star ratings. Empathy is built from (a) catalog structure as a proxy for *who was designed for*, and (b) the household situation the product sits in.



Figure 3. Empathy context: Netflix is often a shared living-room object. A mature-skewed catalog and a family co-viewing job are in tension.

Stakeholder	Job to be done	What this catalog <i>can</i> show	What it <i>cannot</i> show
Household / family viewer	Find something everyone can watch tonight	Share of family-adjacent ratings; children's tags	Whether families watch those titles
16–34 individual subscriber	Personalized mature drama, anime, reality	Density of TV-MA / TV-14; niche tags	Completion, binge depth, churn
Regional commissioning	Local-for-global titles that travel	Production-country credits; type × country	Cost, authenticity, export lift
Content finance	Where mix is thin <i>and</i> valuable	Inventory thickness	ROI, first-28-day views
Competitor strategy	Over-index vs a family-first catalog	Maturity mix; add-year volume	Competitor libraries

Empathy insight (evidence-bounded). The rating architecture is built primarily for teen and adult *individual* viewing. That is a product bet. It is also a household-coverage risk against a family-first competitor. The insight is about inventory design, not preference.

3.2 Define — Point of View and How Might We

A POV that cannot be wrong is not a POV.

A multi-generational household needs a catalog they can co-view without a second subscription, because Netflix's 2021 library is engineered around mature individual titles (TV-MA + TV-14 ≈ 61%) while the living-room job is shared attention — and this dataset can evidence the supply side of that tension, not the demand side.

Secondary POVs: a regional commissioner needs unknown production credits visible (831 titles, 9.4%); a personalization PM needs genre treated as many-to-many (mean 2.19 tags per title).

ID	How Might We	Analytic translation
HMW-1	See growth as policy, not vibe?	<code>date_added</code> vs <code>release_year</code> , split by <code>type</code>
HMW-2	See geographic concentration including unknown production?	Exploded credits + explicit unknown bucket
HMW-3	See maturity mix as strategic position?	Rating; rating × year; rating × genre
HMW-4	Respect multi-genre reality?	Genre-credit grain <code>data/processed/netflix_title_genres.csv</code>
HMW-5	Stop claiming “users prefer X”?	Non-claims on every insight card

Research questions (falsifiable)

- **RQ1.** Did additions (`date_added`) accelerate after mid-2010s expansion, and is 2021 complete?
- **RQ2.** How concentrated are production credits, and how sensitive is the ranking to missing `country`?
- **RQ3.** Is the catalog maturity-skewed, and did that skew hold across add-years?
- **RQ4.** Which genre tags dominate, and which co-locate with mature ratings?
- **RQ5.** Is “family content is scarce” a catalog fact, a mapping artifact, or both?

3.3 Ideate — what we kept and killed

Kept: add-year vs release-year; type split; exploded countries; genre explode; rating × genre as a *cross-tab*; cast-token volume as a production-style signal.

Killed: “teens prefer anime” from a TV-14 × Anime cell; ROI/LTV dashboards (no measures); mean-imputation (no meaningful continuous field); treating TV-show director as MCAR; word clouds; competitor ARPU as a *finding*.

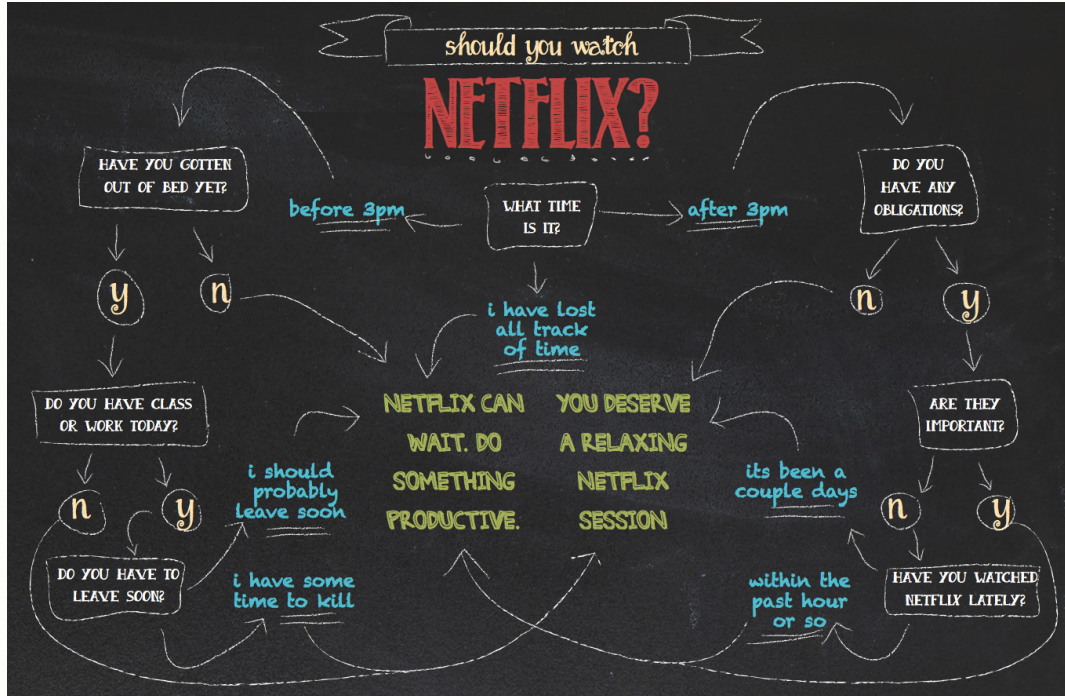


Figure 4. Empathy object (time-use decision). Not a data-generating process for the catalog file.

3.4 Prototype

d.school: a prototype is anything people can react to. Here it is a **decision dashboard**, not a recommender.

Layer	Grain	File
Title catalog (analytic)	one <code>show_id</code>	data/processed/netflix_titles.csv (NULLs preserved)
Title catalog (BI)	one <code>show_id</code>	data/processed/netflix_titles_tableau.csv (English Unknown)
Genre credit	(<code>show_id</code> , tag)	data/processed/netflix_title_genres.csv (19,323 rows)
Country credit	(<code>show_id</code> , country)	data/processed/netflix_title_countries.csv (10,012 rows)

Filters in the workbook: type, rating, listed_in, country, release_year, year of date_added. A viewer should be able to **falsify a headline** by changing windows.

3.5 Test

Test	Studio prototype	After this pipeline
Peak in 2021?	Over-claims a truncated year	Closed. Peak add-year is 2019; 2021 marked right-censored.
Rank countries without Unknown?	Không xác định in Top 10	Closed in Python (831 titles excluded). Tableau PNGs still show the old extract.
rating as quality?	Three Louis C.K. runtimes in rating	Closed. Recoded; rating_was_duration; tests lock it.
Derived band recovers “family”?	TV-G dumped into Other in some studio maps	Closed. Published codebook; family_adjacent = 23.4%.
User segmentation by age?	Still fail — correctly	No viewer ages exist. Keep calling it maturity_band.

4. Data, provenance, and methods

4.1 Source

Item	Value
Corpus	Bansal, S. (2021). <i>Netflix Movies and TV Shows</i> . Kaggle. Originally compiled by Flixable, not an official Netflix dump.
Unit of analysis (title layer)	Title (<code>show_id</code>), <i>not</i> a user, session, or country-year
<i>N</i>	8,807 titles · Movie 6,131 (69.6%) · TV Show 2,676 (30.4%)
<code>release_year</code>	1925–2021
<code>date_added</code>	2008-01-01 → 2021-09-25 · 10 titles missing date added
Duplicate <code>show_id</code>	0
Duplicate <code>title</code>	1 string (Consequences, 2014, Turkey) appears on two <code>show_ids</code> with identical director, cast, runtime, and description — a duplicate record, not a remake

This is observational catalog metadata. It is not a probability sample of global film/TV and not a complete 2021 vintage.

4.2 Data profiling (raw extract; NULLs preserved)

Profiling is performed on the **raw** file, before any display fill. Completeness is $(1 - n_{\{\mathrm{null}\}}/N)$. Distinctness is $(n_{\{\mathrm{unique}\}}/N)$. Filling “No information” and then reporting 100% completeness is not profiling; it is hiding missingness.

Field	Nulls	Completeness	Unique (non-null)	Distinctness	Role
<code>show_id</code>	0	100%	8,807	100%	Primary key
<code>type</code>	0	100%	2	0.023%	Binary product form
<code>title</code>	0	100%	8,806	99.99%	Display name
<code>director</code>	2,634	70.09%	4,528	51.4%	Person credit
<code>cast</code>	825	90.63%	7,692	87.3%	Person credits
<code>country</code>	831	90.56%	748	8.49%	Production-credit <i>combinations</i>
<code>date_added</code>	10	99.89%	1,767	20.1%	Catalog-entry date (string)
<code>release_year</code>	0	100%	74	0.84%	Original year, 1925–2021
<code>rating</code>	4	99.95%	17	0.19%	Includes three leaked durations
<code>duration</code>	3	99.97%	220	2.50%	Heterogeneous units
<code>listed_in</code>	0	100%	514	5.84%	Tag <i>combinations</i>
<code>description</code>	0	100%	8,775	99.64%	Synopsis

Attribute remarks (construct-correct).

- **show_id.** Unique key. String lengths are s1...s8807 (2–5 characters: 9 / 90 / 900 / 7,808 rows). Format variation is padding, not a quality defect.
- **type.** Movie 6,131 (69.62%); TV Show 2,676 (30.38%). Low distinctness is expected for a binary classifier; it is not “poor uniqueness.”
- **title.** One duplicated *string*, *Consequences* (2014, Turkey), on two IDs with identical director, cast, runtime, and synopsis — a duplicate record, not a remake.
- **director.** Completeness 70.09% is the finding. Top frequencies (Rajiv Chilaka 19; Raúl Campos, Jan Suter 18) are title-counts, not “directorial impact.” 91.4% of TV Shows have no director credit.
- **cast.** 825 nulls. Cardinality of the *unexploded* string is 7,692; that is not 7,692 unique actors.
- **country.** 748 unique *strings*, including co-production lists such as “France, Spain”. After explode: 122 country credits and 10,012 rows. The United States / India / UK ranking in Section 4.6 is at exploded grain.
- **date_added.** Object strings with leading spaces; 10 nulls remain null. Modal add-date: 1 January 2020 (109 titles) — a batch update, not a cinematic premiere. Last date: 25 September 2021.
- **release_year.** 74 years, 1925–2021. 2018 is the modal original year (1,147 titles). 57.5% of titles have *release_year* in 2016–2020. That is library *age*, not add policy.
- **rating.** 17 non-null labels in the raw file, **including** 66 min, 74 min, 84 min. Those three are runtimes, not age bands. After recode, *rating* is a maturity label, not an audience review score.
- **duration.** 220 unique strings; movies in minutes, shows in seasons. Never average. The three null durations are the same Louis C.K. rows whose runtime sat in *rating*.
- **listed_in.** 514 combinations; 42 tags after split. Mean 2.19 tags/title.
- **description.** 32 duplicate synopses. Unused as a KPI.

There is **no float column to mean-impute**. The studio PDF’s first preprocessing rule does not apply to this schema.

4.3 Dictionary (construct-correct)

Field	Role	Notes a professional actually needs
show_id	Primary key	s1... in this extract; do not assume stability across Kaggle versions
type	Product form	{Movie, TV Show} only
title	Display name	Not a key
director	Person credit, comma-separated	29.9% missing overall; 91.4% missing for TV Shows
cast	Person credits	9.4% missing; explode → person–title edge list
country	Production credits , comma-separated	9.4% missing; 1,315 multi-country titles; not “filmed in”
date_added	Netflix availability date	Leading spaces; strip before parse. Not cinematic release
release_year	Original release year	Can predate Netflix by decades (library licensing)
rating	Maturity label	MPAA / TV Parental Guidelines-style. Not a crowd score
duration	Heterogeneous	Movies: N min . Shows: N Season(s) . Never average them
listed_in	Genre tags	42 tags after split; 1–3 per title; 514 raw combinations
description	Synopsis	32 duplicate texts; unused as a KPI

4.4 Missingness (kept as NULL)

Field	Nulls	%	Pattern
director	2,634	29.91	Almost all TV Shows
cast	825	9.37	TV Shows 13.1% vs Movies 7.7%
country	831	9.44	TV Shows 14.6% vs Movies 7.2%
date_added	10	0.11	Flag; do not impute
rating	4 raw + 3 duration-swap	0.05 / 0.08	Three Louis C.K. rows recoded

Professional stance. Filling `director` / `cast` / `country` with “No information” is acceptable **only as a dashboard display**. It is unacceptable as statistical imputation, and it makes a profiler report “100% complete” by hiding the signal. Analytic tables keep `NULL`. The Tableau extract uses English `Unknown` at the viz layer only.

The studio PDF filled object columns with “Unknown” / “No information” and claimed post-clean completeness of 100%. That procedure is rejected here. There is also **no numeric float to mean-impute**; `duration` is typed text.

4.5 Recodes and derived grains

1. Three rows with `rating` ∈ {66 min, 74 min, 84 min} (Louis C.K. specials) → runtime restored to `duration`, `rating` set null, `rating_was_duration` = True.
2. `duration` parsed to `duration_value` + `duration_unit` (minutes | seasons).
3. `date_added` stripped → `year_added`; 2021 treated as right-censored.
4. `listed_in` exploded to 19,323 genre-credit rows (mean 2.19 tags/title). Counts are **tag incidences**.

5. `country` exploded to 10,012 country-credit rows (122 distinct credits). Missing country is **not** exploded as a nation.

6. Tests in `tests/test_clean.py` lock these invariants (10 passing).

maturity_band	Ratings	Titles
Kids	TV-Y, TV-Y7, TV-Y7-FV, G, TV-G	908
Teens	PG, TV-PG, TV-14	3,310
Adults	PG-13, R, NC-17, TV-MA	4,499
Unrated	NR, UR	83
Unknown	null, including three recoded duration-swap rows	7

`family_adjacent` = Kids plus PG / TV-PG (**excludes TV-14**). The studio PDF grouped PG-13 with teens; we do not. PG-13 is an MPAA theatrical band whose content is not household-default.

4.6 Robustness: Unknown as a false country (RQ2)

Rank	Including Unknown as a nation	Credits	Excluding Unknown (publication)	Credits
1	United States	3,690	United States	3,690
2	India	1,046	India	1,046
3	Unknown	831	United Kingdom	806
4	United Kingdom	806	Canada	445
5	Canada	445	France	393

The studio Top 10 chart ranks `Không xác định` third. That is a processing threat to validity, not a geographic finding. Figure 9 (Python panel) excludes those 831 titles and states the exclusion in the subtitle.

5. Visualization grammar (shared)

Python figures sit next to Tableau because they share tokens sampled from the workbook PNGs.

Token	Hex	Source in the prototype
Canvas	#EFE8E8	Worksheet cream
Movie	#E15658	Movie stack, Top 10 producing countries
TV Show	#F28B25	TV Show stack, same sheet
Adults	#4471A1	Tableau blue (cast × listed_in)
Teens	#F1CF69	Tableau gold
Kids	#59A14F	Tableau green

Rules: title is a sentence; kicker names the study or the Tableau sheet; 2021 is gray when it is a bar; spaghetti is muted; heatmap is a cross-tab on a cream–coral sequential; no dual axis, no 3D. Full spec: [docs/VIZ_SPECS.md](#).

Data abstraction

Attribute	Type	Grain
release_year	O (year)	title
year_added	O (year)	title (date_added parsed; 10 nulls)
type	N	title
Count of titles	Q	title

Task abstraction (Munzner). Analyze → Consume → **Present** a trend; Search → **Explore** (no lookup of a single title); Query → **Summarize** counts by year × type. Action–target: *present trends* and *compare* Movie vs TV Show.

Idiom. Stacked area (library age) and bars (add-year). Mark: area / bar. Channels: O → horizontal position; Q → vertical position (aligned, highest ranked perceptual task); N → hue (Movie #E15658, TV #F28B25).

Expressiveness / effectiveness. Position on a common scale expresses Q without the pie-chart angle error the studio PDF accidentally described for this task. Color is used only for a two-level categorical split (separable). 2021 on the add-year chart is encoded in gray and annotated as right-censored so the idiom does not *express* a collapse the data cannot support.

TABLEAU

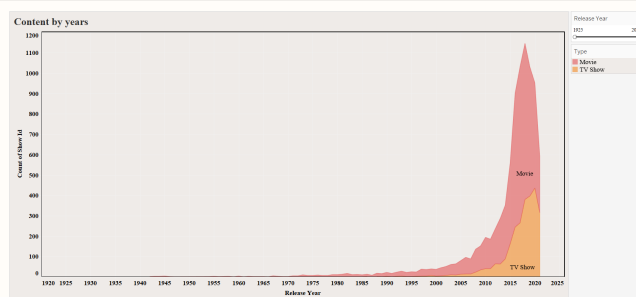
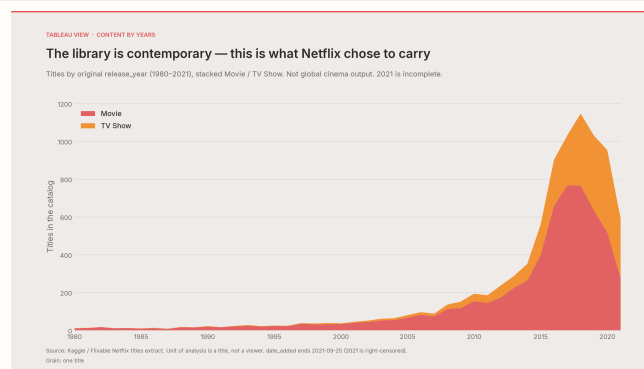


Tableau · Content by years (release_year)

PYTHON



Python · same clock, Movie / TV Show stack

Figure 7. Dual evidence for library age.

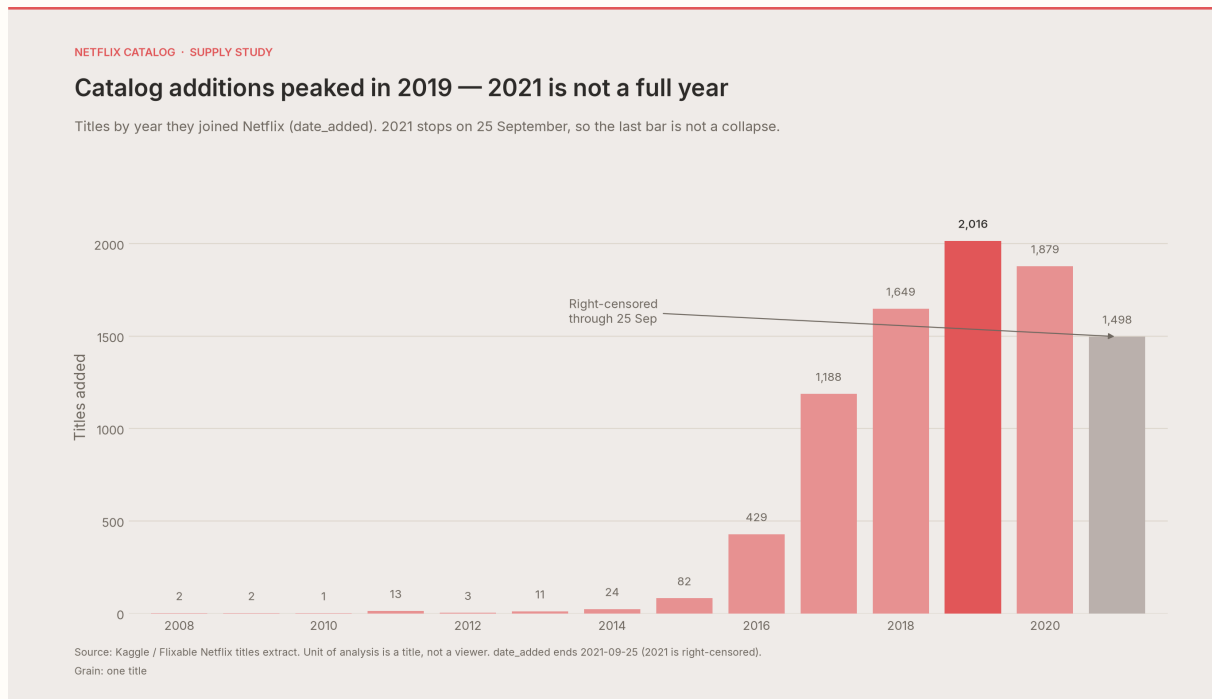


Figure 8. Python-only companion: the commissioning clock (`date_added`) the Tableau sheet did not isolate as a sentence title. Peak = 2019 (2,016). 2021 = 1,498 through 25 September.

Observation. `date_added` peaks in 2019. `release_year` is contemporary-heavy with a long left tail of licensed older films. Movies dominate the stack in both clocks; TV Shows rise after the mid-2010s but never overtake.

Inference. Do not read “content by years” as cinema output. It is what Netflix chose to carry. Add-year is expansion tempo.

Non-claim. Not global production volume. 2021 cannot be compared to 2019 without an annualization this file does not support.

Year of <code>date_added</code>	Titles added
2016	429
2017	1,188
2018	1,649
2019	2,016
2020	1,879
2021 (through 25 Sep)	1,498

Task 2 — Production geography (RQ2, HMW-2)

Why. Concentration is a strategy claim only if missingness is visible.

Data abstraction. `country_credit` (N, exploded); `type` (N); count of credits (Q). Grain = one (`show_id`, country). A US–India co-production is two rows. 831 titles with null country contribute **zero** rows in the publication ranking.

Task. Present ranking; explore concentration; summarize Top 10. Choropleth: present spatial distribution (Q → color on a map; location → position).

Idiom. Horizontal stacked bars (ranking) + choropleth (geography). Ranking uses aligned length — appropriate for Top 10 comparison. A map is the right form for *where*, but a poor form for *how much* (area and projection distort); hence both idioms.

Studio failure. Ranking Unknown / không xác định as country #3. Python excludes it and writes the 9.4% into the subtitle.

TABLEAU

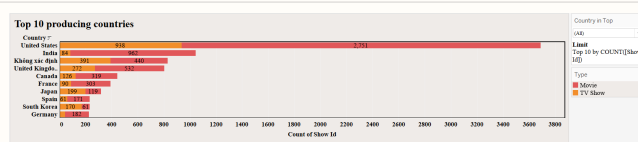
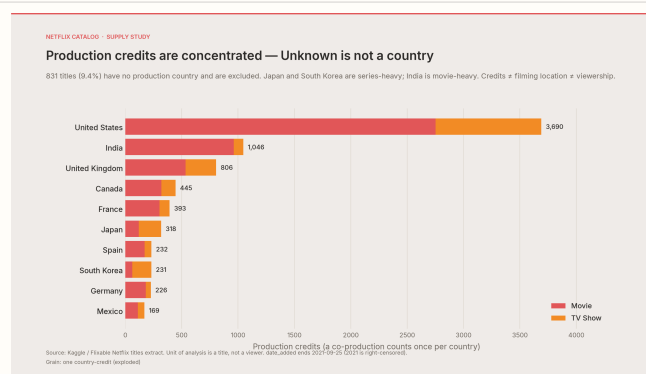


Tableau · Top 10 (Unknown still in the extract)

PYTHON



Python · Unknown excluded (831 titles / 9.4%)

Figure 9. Dual evidence for production-credit ranking.

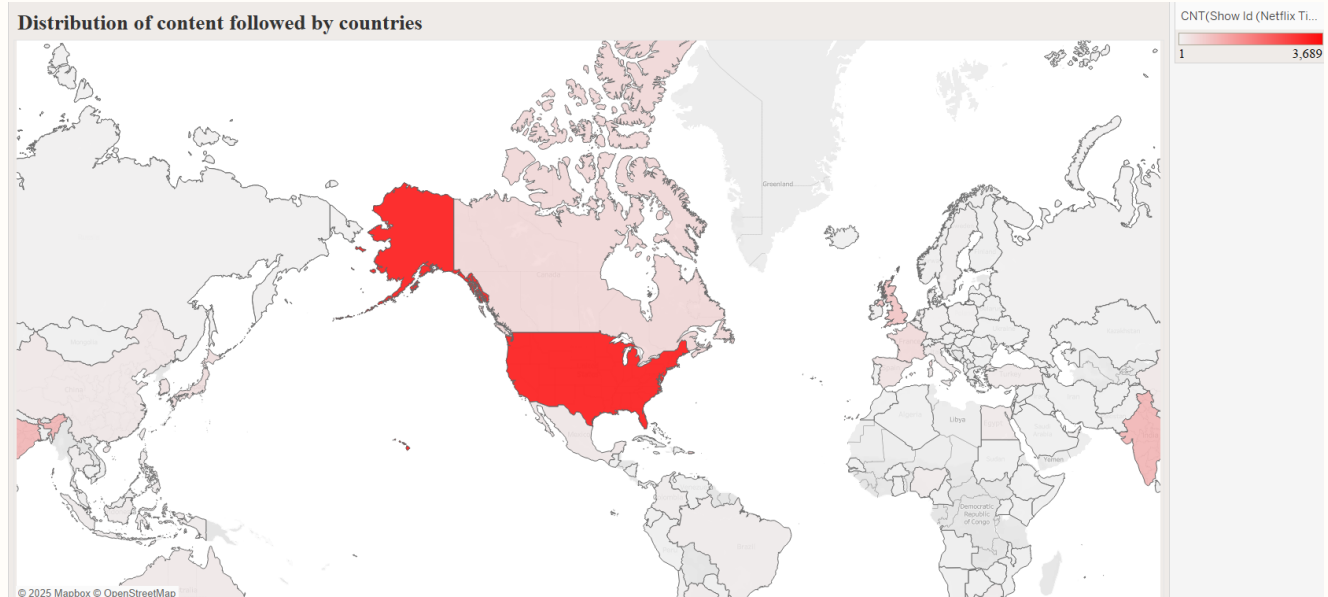


Figure 10. Tableau-only companion: choropleth of production credits. The map is the right geographic idiom; Figure 9 is the honest ranking. Color is a sequential red (not rainbow).

Observation. United States 3,690 · India 1,046 · United Kingdom 806. If missing country is promoted to a nation label, **Unknown ranks third (831)**. Japan and South Korea are series-heavy; India is movie-heavy.

Inference. Local-for-global is a long tail of credits, not a dethroning of the U.S.

Non-claim. Credit $\neq$ filming location $\neq$ the market that watched it. The studio dictionary's “countries where the movie was filmed” is false on this field.

Task 3 — Genre tags over time (RQ4, HMW-4)

Data abstraction. genre (N, 42 tags); year_added (O); incidences (Q). Grain = one (show_id, tag). International Movies = 2,752 **tag incidences**, not 2,752 exclusive titles.

Task. Explore trends; summarize top tags; present a readable series (not 42 equally loud lines).

Idiom. Polylines. Channel: O $\rightarrow$ x, Q $\rightarrow$ y, N $\rightarrow$ color **for five highlights**; remaining series in muted gray (#D5D0C8). The studio spaghetti uses fully saturated color for every tag (low discriminability). Publication mutes the long tail — an effectiveness fix, not a data change.

TABLEAU

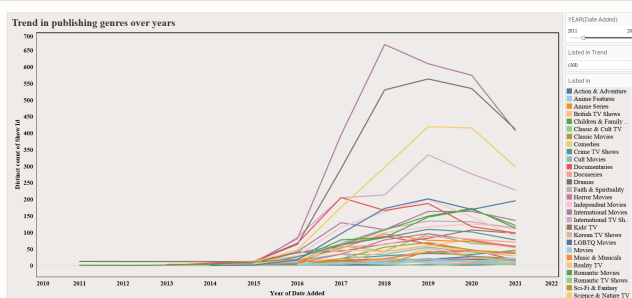
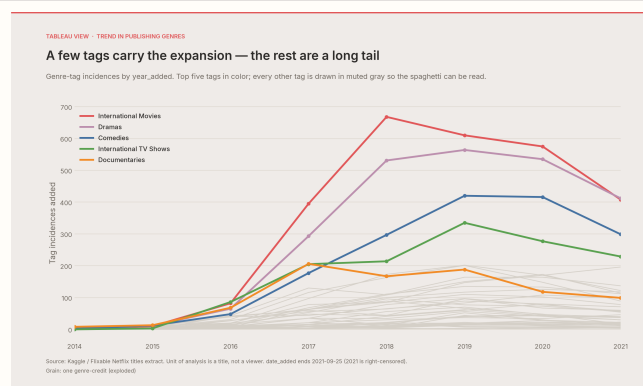


Tableau · Trend in publishing genres

PYTHON



Python · top five in color, long tail muted

Figure 11. Dual evidence for genre-tag trends (exploded grain).

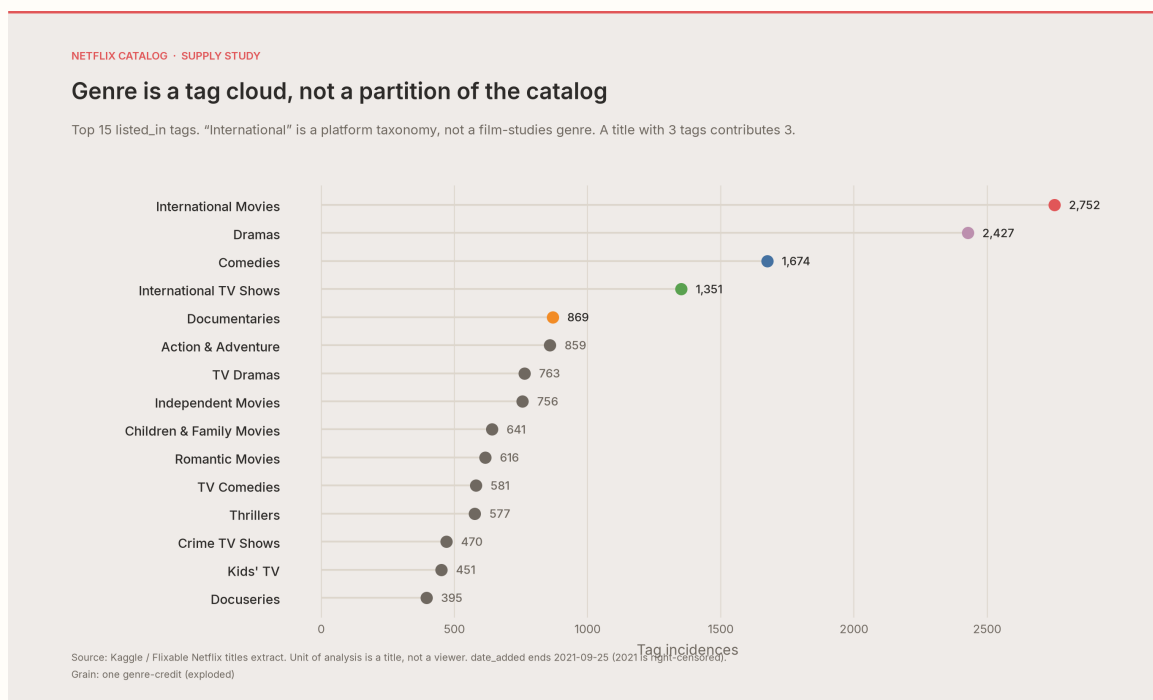


Figure 12. Python ranking of tag incidences (Top 15). International Movies 2,752 · Dramas 2,427 · Comedies 1,674.

Observation. A few tags carry the 2016–2019 expansion; the rest are a long tail. Mean tags per title = 2.19.

Inference. “International” is a platform taxonomy. Drama + comedy are the mass spine.

Non-claim. Tag growth ≠ audience growth.

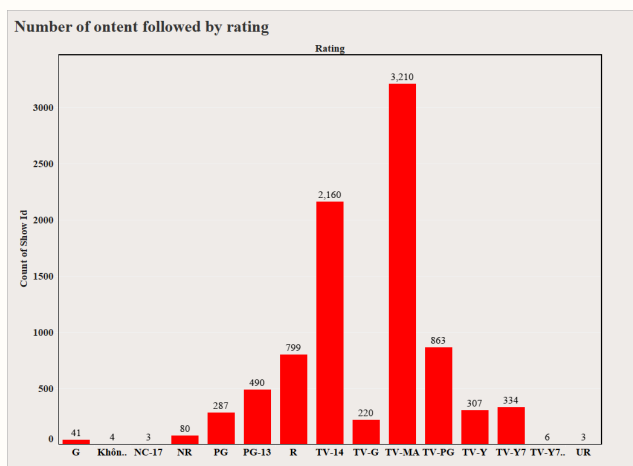
Task 4 — Maturity mix (RQ3, RQ5, HMW-3)

Data abstraction. rating (N, ordered for display along the parental-guideline ladder); maturity_band (N, derived); count of titles (Q). Grain = title.

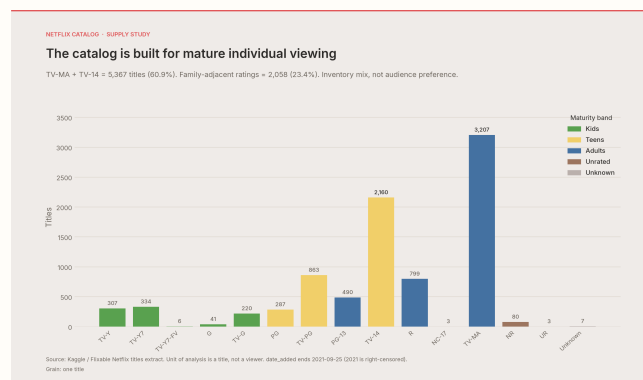
Task. Present distribution; compare bands; summarize family-adjacent share.

Idiom. Vertical bars. Q → aligned bar height (accurate); band → hue (Kids green, Teens gold, Adults blue). The studio sheet uses a single red for every rating, which **fails to express the band grouping** the analysis then claims in prose.

TABLEAU



PYTHON



Python · recoded ratings, maturity bands

Tableau · Number of content by rating

Figure 13. Dual evidence for title-level maturity mix.

Observation. TV-MA 3,207 · TV-14 2,160. Family-adjacent = 2,058 (23.4%). Adults 4,499 + Teens 3,310 dominate; Kids 908.

Inference. The 2021 catalog is positioned as a teen/adult destination. Disney+ contrast is a *positioning* claim, not a measurement of Disney’s library.

Non-claim. “Users are mostly adults” does not follow from inventory mix. rating is not a quality score. The three 66/74/84 min “ratings” in the studio chart are recoded runtimes.

Task 5 — Maturity over time (RQ3)

Data abstraction. release_year or year_added (O); maturity_band (N); count (Q). Grain = title.

Task. Present composition over time; explore whether the mature skew is recent.

Idiom. Stacked area / stacked bars. Same band hues as Task 4 for **identity consistency** across the paper (a grammar the studio line chart of 18 ratings violates).

TABLEAU

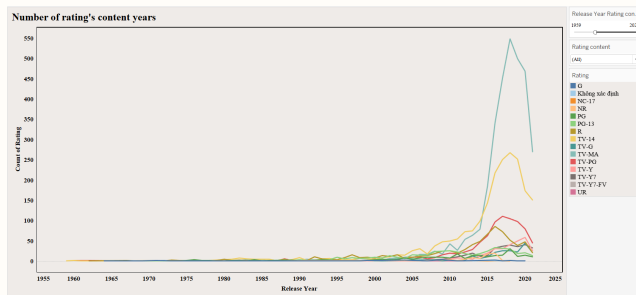
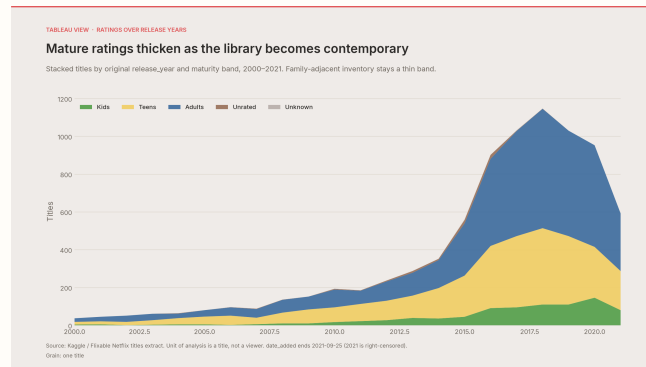


Tableau · Ratings over release years (18 series)

PYTHON



Python · five bands, 2000–2021

Figure 14. Dual evidence for maturity over original release year.

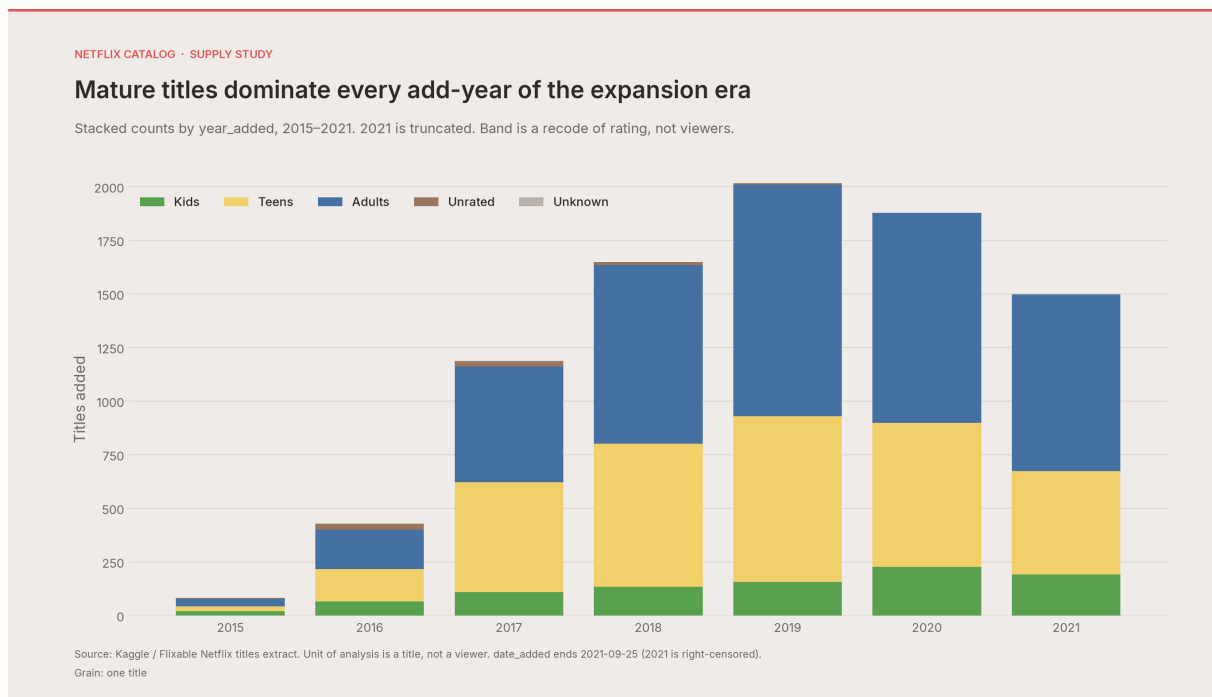


Figure 15. Python companion on the add-year clock (2015–2021). Mature bands dominate every expansion year.

Observation. Family-adjacent inventory stays a thin band as the library becomes contemporary.

Inference. The mature position is not an accident of a single year.

Non-claim. Not “the audience got older.”

Task 6 — Rating × genre is a cross-tab, not a correlation (RQ4)

Data abstraction. rating (N); genre (N); cell = count of (show_id, tag) pairs (Q).

Task. Present a matrix; explore which cells dominate; summarize modal inventory cell.

Idiom. Heatmap. Two categorical positions + Q → sequential lightness/hue. Named **cross-tab**. The studio filename says “correlations”; **no coefficient is computed**. Color is cream → coral → Movie red, not rainbow (rainbow is not monotonic; it fails expressiveness for Q).

TABLEAU

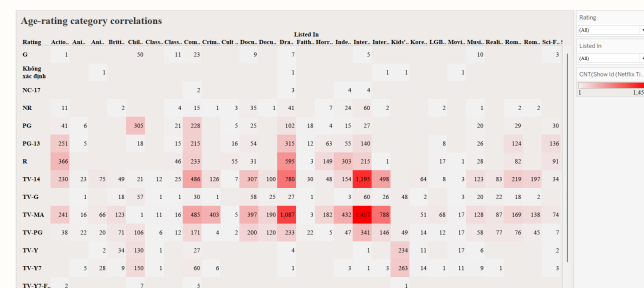
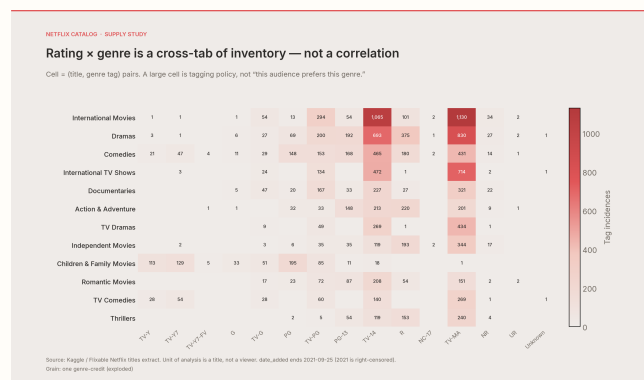


Tableau · filename claims “correlations”

PYTHON



Python · named as a cross-tab

Figure 16. Dual evidence for rating × genre. Grain = genre-credit.

Observation. TV-MA × International Movies = 1,130 incidences; TV-14 × International Movies = 1,065.

Inference. Mature international drama is the modal *inventory* cell.

Non-claim. Cell size is tagging policy, not taste affinity.

Task 7 — Cast name-tokens by genre

Data abstraction. Sum of parsed cast-list lengths by genre tag (Q); genre (N). Grain = (genre tag × cast name-token). 825 titles with missing cast are excluded from the sum.

Task. Present a ranking of accumulated names; explore format (ensemble vs lean).

Idiom. Horizontal bars. Q → length. Color reuses the genre highlight set for consistency, not a 42-hue rainbow.

TABLEAU

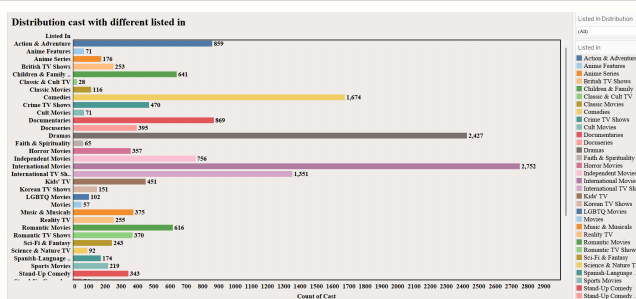
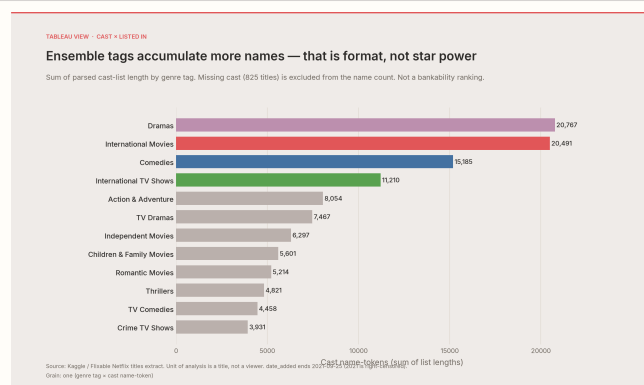


Tableau · Cast × listed in

PYTHON



Python · name-tokens; missing cast excluded

Figure 17. Dual evidence for cast-token volume by genre tag.

Observation. Dramas and International Movies accumulate the most parsed names.

Inference. That is **format** (ensemble drama) more than star power.

Non-claim. Not a talent graph or bankability ranking until names are resolved and credited uniquely.

Task 8 — Missingness as a KPI (Python-only diagnostic)

The course workbook never isolated missingness as a view. Test required it.

Data abstraction. Field $\in$ {director, country, cast} (N); type (N); percent null (Q, 0–100). Grain = title.

Idiom. Grouped horizontal bars. Q $\rightarrow$ length (so 91.4% vs 3.1% is readable to a percentage point).

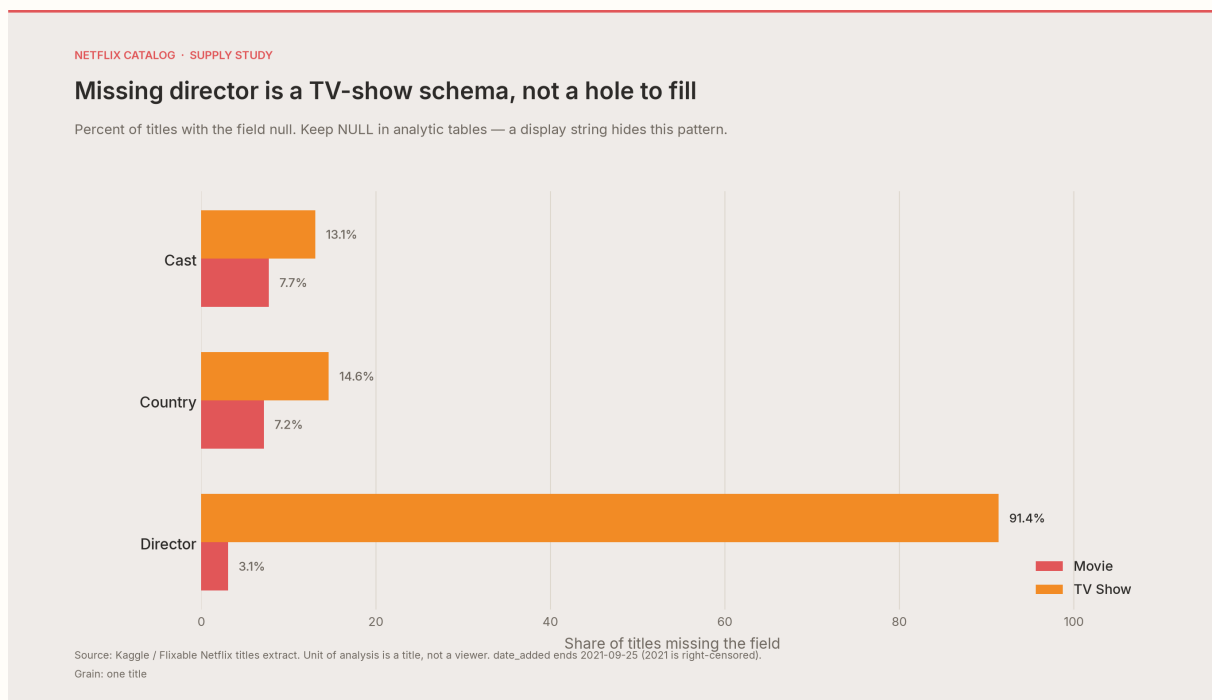


Figure 18. Python-only: percent missing by field and type. No Tableau pair — the studio extract filled blanks and then profiled “100% complete.”

Observation. Director missing for 91.4% of TV Shows vs 3.1% of Movies.

Inference. TV-show director is a schema/practice difference, not a hole to fill.

Non-claim. Completeness of metadata is not a quality score for the title.

7. Synthesis

Design Thinking is supposed to end in a **sharper problem**, not a longer slide of recommendations.

7.1 What survived Test

1. The catalog is a designed **maturity position**. TV-MA + TV-14 dominance is too large to be a tagging accident.
2. The growth era is **dated**. Additions scale from 2016, peak in 2019, and cannot be scored for full-year 2021.
3. Geography is a **hub-and-spoke credit structure** with a material unknown class (9.4%) that, if promoted, ranks third.
4. Genre must be modeled as **multi-label**. Any “top genre” chart is an incidence chart.
5. Family coverage is a **supply** question this file can answer; family *demand* is not.

7.2 Implications (scoped to people who own catalogs)

Horizon	Move	Why this file supports it	What you must still measure
Immediate	Missingness dashboards (country, director by type)	Missingness is large and structured	Nothing — this is free
Immediate	Stop ranking Unknown as a country	831 rows	—
Near	Family-adjacent coverage monitor	23.4% family-adjacent ratings	Household accounts, kids profiles, co-viewing
Near	Treat India / Korea / Japan as format-different poles	Type × country stacks	Cost per hour, export hours watched
Later	Join even a 1% viewing sample before ROI language	Current file cannot	Views, completion, retention, cost



Figure 19. Competitive landscape is context for Empathize. Subscriber and ARPU figures are not estimated here.

8. What this study does not claim

A 10/10 data study is defined as much by **refused inferences** as by charts.

Tempting sentence	Why it is invalid on this file
“Teens prefer anime / adults prefer drama”	No viewers, no ages. maturity_band is a title-level recode of rating
“Netflix should cut volume and chase ROI”	No cost, no hours viewed, no LTV
“Korea is winning because K-content is popular”	Title credits, not hours watched, and not post- <i>Squid Game</i> 2021–22 demand
“Disney+ beats Netflix with families”	Disney’s catalog is not in the data
“Ratings measure quality”	rating is a maturity label; three duration-swap rows are recoded
“The map is where titles were filmed”	Production credits, often co-production
“2021 was a collapse year”	Right-censored at 25 September

9. Threats to validity

Threat	How it shows up	Mitigation
Construct	Tags as taste; <code>rating</code> as quality; <code>maturity_band</code> as viewers	Section 8; codebook; tests
Internal	Reading <code>release_year</code> as Netflix strategy	Prefer <code>date_added</code> as the policy clock
External	Snapshot ends 2021-09-25; no ad-tier, games, or 2022–2026 originals	Bound every claim to “in this extract”
Measurement	Multi-label and multi-country explode inflate counts	Always state grain
Processing	Localized fill-ins leak into Top 10	Analytic NULL; Python ranking excludes Unknown
Selection	Kaggle/Flixable ≠ Netflix title master; regional catalogs differ	Do not treat $N = 8,807$ as “the global product”
Record identity	Two <code>show_ids</code> share one <code>Consequences</code> record	Flagged; not treated as two titles in narrative counts of “unique works”

10. Reproducibility

```
python -m pip install -r requirements.txt
python -m pytest -q          # 10 wrangling invariants
python -m netflix_catalog    # gold tables + quality report + figures/viz_*.png
```

Artifact	Path
Analytic titles	data/processed/netflix_titles.csv
Tableau extract	data/processed/netflix_titles_tableau.csv
Quality report	docs/DATA_QUALITY.md
Dictionary	docs/DATA_DICTIONARY.md
Viz spec	docs/VIZ_SPECS.md
Workbook	Tableau/Netflix & TV Show.twbx

Open the prototype in Tableau Desktop / Tableau Public. Point a refresh at `data/processed/netflix_titles_tableau.csv`. Alias `Unknown` as “No production country” and keep it out of Top 10.

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Appendix A — Errata of the original studio PDF

Original claim	Correction
<code>country</code> = filming location	Production credits, often multi-valued
<code>rating</code> = audience reviews	Parental-guideline / MPAA-style maturity label
<code>date_added</code> = released on Netflix as cinematic release	Catalog-entry date
Mean-impute numeric blanks	No meaningful float; <code>duration</code> is typed text
Fill objects then report 100% completeness	Hides MNAR/structural missingness
<code>Không xác định</code> as a Top-10 country	831 missing credits; exclude from ranking
<code>66 min / 74 min / 84 min</code> as ratings	Duration leaked into <code>rating</code> (Louis C.K.); recoded
Heart-disease / gender / cholesterol sentences in idiom evaluation	Copy-paste from the assignment example; deleted
Pie-chart evaluation attached to a bar idiom	Idiom and evaluation now match
Duplicate title = remake	Two identical <code>Consequences</code> records (same credits and synopsis)
Teen band includes <code>PG-13</code>	<code>PG-13</code> coded Adults; <code>family_adjacent</code> excludes <code>TV-14</code>
Map idiom described as “line chart”	Choropleth; Q → sequential color

Appendix B — Figure index (Tableau ↔ Python)

Fig.	Tableau export	Python publication
5–6	Dashboard.png	<code>viz_08_briefing_board.png</code>
7–8	Content by years.png	<code>viz_09_library_by_release_year.png</code> + <code>viz_01_catalog_additions.png</code>
9–10	Top 10 producing countries.png + choropleth	<code>viz_04_producing_countries.png</code>
11–12	Trend in publishing over years.png	<code>viz_10_genre_trends.png</code> + <code>viz_05_genre_incidence.png</code>
13	Number of Content followed by rating.png	<code>viz_03_maturity_mix.png</code>
14–15	Number of rating's content years.png	<code>viz_11_rating_over_release_year.png</code> + <code>viz_07_maturity_over_add_year.png</code>
16	Age rating category correlations.png	<code>viz_06_rating_genre_crosstab.png</code>
17	Distribution cast with different listed in.png	<code>viz_12_cast_by_genre.png</code>
18	— (not in workbook)	<code>viz_02_missingness_by_type.png</code>

One-sentence contribution: we specified who the catalog is for, wrote a POV that can be wrong, killed inferences the file cannot support, and left a prototype that makes those limits visible.